

## AIR POLLUTION CONTROL - FIELD INSPECTION WORK SHEET

| Key Punch Data: |         | County<br>Number | Source<br>Number | Emission<br>Point # | Action<br>Type | Date<br>Mo Day Yr | Agency/<br>Indiv. ID |
|-----------------|---------|------------------|------------------|---------------------|----------------|-------------------|----------------------|
| Line 1          | 1011101 | 021201           | 11100131         | 0001                | 1114           | 1011210181        | 10001                |
|                 | 1 3     | 4 7              | 8 12             | 13 15               | 16 18 52       | 54 59             | 66 68 80             |

### Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

$$\begin{array}{r} 1010151 \\ 16 \quad \quad 18 \end{array}$$
$$\begin{array}{r} 131 \\ 78 \overline{) } \end{array}$$
$$\begin{array}{r} 1 \text{ C1} \\ 8 \text{ 0} \end{array}$$

(NOTE: Columns not shown must be left blank)  
( Do not punch Line 2 if column 78 is blank)

Company: Southwestern Portland Cement Co.

Inspected by: G. Mathews

Site Location: Lyons CO

County: Boulder

Contact Person: Randy Wiley, Ed Parker

Time: 0900 - 1300

Telephone No.: 534-4206

EIS Update: *Attached*

Inspection: Routine ☒ Complaint ☐ FA ☐

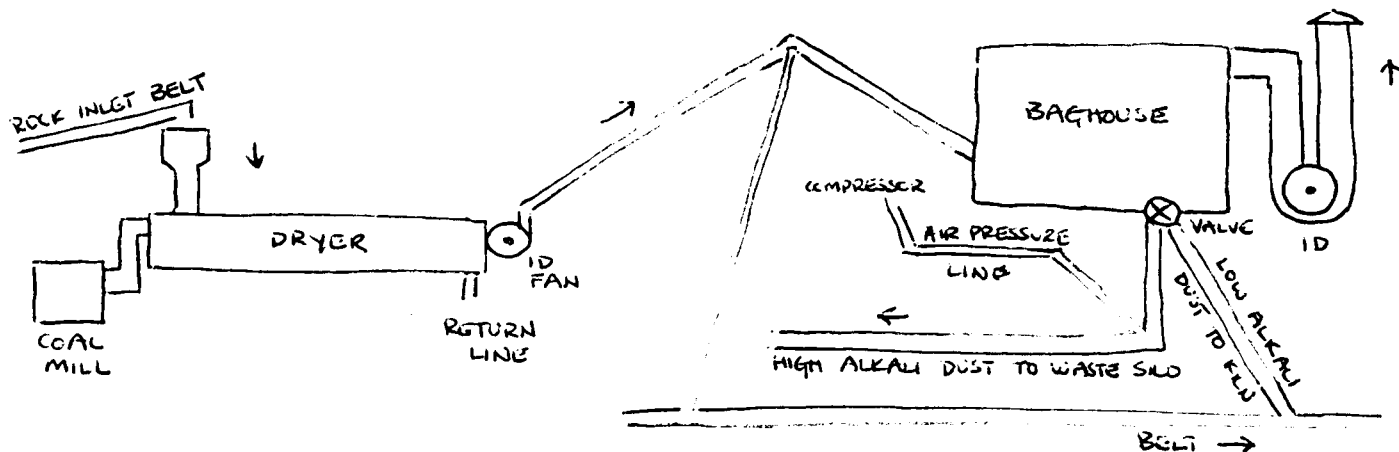
12.75 Hrs.\* Permit No. C-12 444

Emission Points Subject to NSPS: *NA*

| Major source inspection per | 1988 APCD Contract |
|-----------------------------|--------------------|
|                             |                    |

Immediately upon arrival, a tour of the facility was conducted with particular attention to emission points. The following narrative relates site conditions at the time of the inspection.

EIS Pt 06. <sup>Limestone dryer</sup> At the time of inspection, the exhaust point was producing a significant steam plume. It was hypothesized that due to the precipitation the previous week, the moisture content of the raw material rock was high. The facility was burning gas at the time. There was a noticeable dust buildup on the pad below and around the dryer.



p1 of , continued

Southwestern Portland Cement Co, Source No. 3  
9-20-88 inspection report continued.

The dryer is permitted by C-12,444 (1), modification 4-8-85, which is Initial Approval only, no Final Approval.

During 1987, a total of 521,452 tons of limestone were crushed and dried, which is less than the 717,000 tons limit under Condition 3 of the Permit. The dryer burned 41.885 million ft<sup>3</sup> of natural gas (Permit limit is 50 million ft<sup>3</sup>) and 2,218 tons of coal (Permit limit is 7,446 tons).

At 1205, the pressure differential across the dryer baghouse was 6.5 psi, the combustion temperature was 1650°F. The dryer feed rate was 120 tons/hr.

EIS Pt. #2 Materials crushing      Opacity observations were made at the following locations:

|                            |    |
|----------------------------|----|
| Primary crusher baghouse   | 0% |
| Surge tank (silo) baghouse | 0% |
| Waste dust silo baghouse   | 0% |

The waste dust silo truck loading bay was not in use at the time of inspection. Fugitive emissions have been a problem during past inspections. At this time the dust control plastic shrouding around the bay was badly torn.

Tanker trucks were being loaded with waste dust at the time. High opacity fugitive emissions were observed at the tanker manholes. Opacity readings of this intermittent source were taken.

|        | 00   | 15   | 30   | 45   |
|--------|------|------|------|------|
| 1048   | —    | —    | —    | —    |
| * 1049 | —    | 100% | 100% | 100% |
| 1050   | 100% | 5%   | 5%   | 5%   |
| 1051   | 0%   | 100% | 100% | 100% |

, spillage, truck departure

This point needs follow-up inspection.

Southwestern Portland Cement Co., Source No. 3  
9-20-88 inspection report continued

EIS Pt 09 Kiln Stack . Opacity was 0% at the time of inspection. At 1205, the control room indicated that the kiln feed rate was 90 tons/hr, the cement production rate was 60 tons/hr. The feed rate was stated to be normally about 103 tons/hr. The pressure differential across the kiln baghouse was 2.649 psi; across the alkali baghouse 1.5 psi

EIS Pt 07 Homogenizing Silo . Opacity was 0%.

EIS Pt 01 Clinker cooler . Opacity from the exhaust stack was 0%.

\* Fugitive dust emissions are still a problem at the hot clinker bay. This open bay creates a fugitive dust haze over the whole plant. SW Portland is considering a hotter temperature belt which would reduce the need for dropping hot clinker so often. Mr. Wiley also thinks that a neg. pressure pull into a baghouse would help. Another source of dust is from the drop chute doors, which were torn at the time of inspection.

EIS Pt. 11 Mill Opacity was 0% at both raw and finished mill exhaust points.  
EIS Pt. 08 A-Frame and outside clinker storage.

No emissions were observed from the A-Frame. At this time the loading hopper on the south side of the A-Frame was not being used. This has been a source of dust in the past.

Outside clinker is permitted by 84 BO 396 F. The dimensions of the only pile, which was immediately west of the A-Frame were approximately 270' long (N-S), 75' wide, 20' high. This is about 15,000 yd<sup>3</sup> of clinker, which will be well below the 120,000 tons limit.

Southwestern Portland Cement Co., Source No. 3

9-20-88 inspection report continued.

EIS Pt 05. Site fugitive dust More dust accumulation was noted during this visit than on past inspections.

EIS Pt 10 Finished products silos. Opacity was 0%.

EIS Pt. 04 Quarries

A-Pit was not working on the day of the inspection. This quarry is usually worked 2-3 times a week. The site haul roads were well watered.

At the E-Pit, south of Hygiene Road, SWPC was filling a gully at the south end of the property with waste rock overburden from the pit. Topsoil was being applied over the waste rock. The topsoil was from a dry stockpile and as the paddle scraper dragged the pile, lots of blowing fugitive dust was being produced.

The south end of the property was traded to Marigold 21.

\* A small drilling rig was operating at E-Pit. The usual rig was in for maintenance. On this small rig the water injection line was broken and fugitive dust was 60%-100%.

Truck hauling crews work 4 @ 10 hr days, Mon - Thur. There are usually 3 trucks operating at 30 trips/truck/day. The maximum trip is 1.0 mile one way. This day the trucks were driving 1/2 - 3/4 round trip. The range is between 25-75 miles/truck/day. This is about 36,000 VMT/year.

A water truck operates 7 days/week 6-10 trips/day. Water only is used as a dust suppressant.

Southwestern Portland Cement Co., Source No. 3  
9.20.88 inspection report continued.

After the facility inspection, the Control Room was visited in order to note present operating parameters and examine recordkeeping.

The kiln feed, production, and dryer rates were noted and are recorded above. Also the control equipment parameters were also noted.

Records are maintained:

production rate, tons/day

kiln feed rate, tons/day

Moisture content 9.20.88, 0300 : 0.21%

0700 : 0.24%

1100 : 0.18%

hours of operation, hrs/day

fabric filter maintenance : preventive maintenance records are kept. The big baghouses are inspected weekly.

start-ups, shut-downs, malfunctions : A log book is kept.

Quarterly operations records for 1987 were received, and SWPC forwarded analysis reports for coal, oil, and gas.

The problem areas noted during the 1987 inspection were reiterated:

hot clinker bay fugitive dust

secondary crusher baghouse

quarry - permits + APENs

dryer - gas limits

NO<sub>x</sub> stack test results

A-Frame - outside hopper fugitive dust.

kiln - stack test

gypsum silo baghouse

Southwestern Portland Cement Co., Source No 3  
4-20-88 inspection report.

SWPC responded to these items in a letter dated  
9-27-88, a copy of which is attached.

I discussed the need for Permits on each of the quarry  
pits w/ Tom Tistic. We concurred that because the  
plans for the pits preceded 1972, Permits will not  
be required by F.D. Control Plans will

Status : In compliance by inspection and calculation.

juu  
5/8/89

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

|                                    |                         |
|------------------------------------|-------------------------|
| SOURCE NUMBER: 3                   | INSPECT TYPE: ANNUAL    |
| SOURCE NAME: SW PORTLAND CEMENT CO | ACTION TYPE: 12         |
| LOCATION: LYONS                    | INSPECTOR: G. MATHEWS   |
| COUNTY: BLDR 0220                  | HRS REQUIRED: 12.75 HRS |
| CONTACT: RANDY WILEY               | PREPARATION: 1.00 HRS   |
| TELEPHONE: 534-4206                | TRAVEL: 1.00 HR         |
| DATE: 9-20-88                      | INSPECTION: 4.00 HRS    |
| TIME: 9:00-1:00                    | WRITE UP: 6.25 HRS      |
|                                    | EIS: 0.50 HRS           |

|                            |                         |
|----------------------------|-------------------------|
| EIS POINT ID: 01           | PERMIT: GRANDFATHERED   |
| POINT NAME: CLINKER COOLER | CONTROL EQUIP: BAGHOUSE |

NSPS REG?: YES: 6,IV,C,2,a & b

NESHAPS REG?: NO

PSD REG?: NO

1987 ANNUAL THRUPUT:

KILN FEED: 493,292 TONS

CEMENT PRODUCED: 352,161 TONS

OP'NAL SCHED:

5061.5 HR/YR 1987

| POLLUTANT                      | PART       | OPACITY    | VOC | CO | NOX | SOX | LEAD |
|--------------------------------|------------|------------|-----|----|-----|-----|------|
| REGULATION                     | 6,IV,C,2,a | 6,IV,C,2,b |     |    |     |     |      |
| DESIGN RATE                    |            |            |     |    |     |     |      |
| HR ALLOW EM                    |            | 10%        |     |    |     |     |      |
| ALLOW EM (LB/TON FEED TO KILN) | 0.1        |            |     |    |     |     |      |

|                               |             |      |
|-------------------------------|-------------|------|
| YR THRUPUT (TONS CEMENT PROD) | 352161      |      |
| REFERENCE                     | 3-05-006-14 | INSP |
| EM FACTOR (LB/TON)            | 9.2         |      |
| CONT FACTOR                   | 0.999       |      |
| YR ACTUAL EM (TONS)           | 1.620       |      |

|                                 |        |
|---------------------------------|--------|
| KILN FEED (TONS)                | 493292 |
| ACTUAL EM (LB/TON FEED TO KILN) | 0.007  |

|              |    |
|--------------|----|
| HR ACTUAL EM | 0% |
|--------------|----|

|                         |    |     |
|-------------------------|----|-----|
| COMPLIANCE (3=YES;1=NO) | 3  | YES |
| REV APEN?               | NO |     |

## COMMENTS:

1. FILE NAME: 3\_1\_88
2. DUMPING OF HOT CLINKER STILL A PROBLEM; SEE INSPECTION NARRATIVE
3. THE AMOUNT OF CEMENT PRODUCED IN A YEAR DOES NOT NECESSARILY COME ONLY FROM THE THE CLINKER (AND THEREFORE THE FEED) THAT WAS PRODUCED IN THAT SAME YEAR. SOME OF THE CEMENT PRODUCED COMES FROM THE CLINKER STOCKPILED IN PREVIOUS YEARS. SO THE RATIO OF LB OF PARTICULATE EMISSIONS TO TONS OF FEED TO THE KILN IS AN ERRONEOUS CALCULATION.

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: UNASSIGNED, 02  
 POINT NAME: RAW MTL PRIMARY CRUSHER  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: P-10,225  
 CONTROL EQUIPMENT: BAGHOUSES

559,886 TONS  
 4062.0 HR/YR 1987

| POLLUTANT               | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|-------------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION              | 1,III,C,1,b | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TONS/HR)   | 200         |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)     | 40.407      | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)       | 559886      |          |     |    |     |     |      |
| REFERENCE               | 3-05-006-09 | INSP     |     |    |     |     |      |
| EM FACTOR (LB/TON)      | 0.5         |          |     |    |     |     |      |
| CONT FACTOR             | 0.999       |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)     | 0.140       |          |     |    |     |     |      |
| TIME FACTOR (HR/YR)     | 4062        |          |     |    |     |     |      |
| HR ACTUAL EM (LB/HR)    | 0.069       | UNKNOWN  |     |    |     |     |      |
| COMPLIANCE (3=YES;1=NO) | 3           | UNKNOWN  |     |    |     |     |      |
| REV APEN?               | NO          |          |     |    |     |     |      |

EIS POINT ID: UNASSIGNED, 02  
 POINT NAME: RAW MTL SECONDARY CRUSHER  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: GRANDFATHERED  
 CONTROL EQUIPMENT: BAGHOUSES

559,886 TONS  
 4062.0 HR/YR 1987

| POLLUTANT             | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|-----------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION            | 1,III,C,1,b | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TONS/HR) | 200         |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)   | 40.407      | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)     | 559886      |          |     |    |     |     |      |
| REFERENCE             | 3-05-006-09 | INSP     |     |    |     |     |      |
| EM FACTOR (LB/TON)    | 1.5         |          |     |    |     |     |      |
| CONT FACTOR           | 0.999       |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)   | 0.420       |          |     |    |     |     |      |
| TIME FACTOR (HR/YR)   | 4736        |          |     |    |     |     |      |
| HR ACTUAL EM (LB/HR)  | 0.177       | 0-5%     |     |    |     |     |      |



COMPLIANCE (3=YES;1=NO) 3 YES  
REV APEN? NO

POINT NAME: RAW MATERIALS TRANSFER PERMIT: P-10,298; P-10,535  
CONTROL EQUIP: TOTALLY ENCLOSED  
W/ NEG PRESSURE TO BAGHOUSE  
1987 ANNUAL THRUPUT: 559,886 TON/YR  
OP'NAL SCHED: 4062.0 HR/YR

| POLLUTANT                 | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|---------------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION                | NO SPEC     | 1,II,A,1 |     |    |     |     |      |
| HR ALLOW EM               |             | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)         | 559886      |          |     |    |     |     |      |
| REFERENCE                 | 3-05-006-12 |          |     |    |     |     |      |
| EM FACTOR (LB/TON)        | 0.3         |          |     |    |     |     |      |
| CONT FACTOR               | 0.999       |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)       | 0.084       |          |     |    |     |     |      |
| HR ACTUAL EM              |             | 0-5%     |     |    |     |     |      |
| COMPLIANCE                | NA          | YES      |     |    |     |     |      |
| YR ACTUAL EM TOTAL (TONS) | 0.644       |          |     |    |     |     |      |

COMMENTS:

1. FILE NAME: 3\_2\_86
2. APCD FORWARDED COPY OF P-10,225 (PRIMARY CRUSHER BAGHOUSE) TO BCHD ON 12-15-88. PRIOR TO THIS DATE, POINT WAS LISTED AS GRANDFATHERED. OTHER PERMITS THAT MIGHT APPLY TO MATERIALS HANDLING ARE P-10,298 (NO. 4 AND NO. 6 BELT CONVEYORS) AND P-10,535 (SURGE HOPPER OF NO. 2 BELT).

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: UNASSIGNED,03  
 POINT NAME: COAL HANDLING  
 NSPS REG?: NO  
 MESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: P-10,718  
 CONTROL EQUIP: BAGHOUSE

COAL BURNED: 48,699 TONS  
 24HRS/7DAY/47WKS/YR=7896HR/YR

| POLLUTANT            | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|----------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION           | 1,III,C,1,a | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TON/HR) | 13          |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)  | 17.609      | 20%      |     |    |     |     |      |

|                      |                     |
|----------------------|---------------------|
| YR THRUPUT (TONS)    | 48699               |
| REFERENCE            | 3-05-010-10 NO INSP |
| EM FACTOR (LB/TON)   | 0.02                |
| CONT FACTOR          | 0.999               |
| YR ACTUAL EM (TONS)  | .000                |
| TIME FACTOR (HR/YR)  | 7896                |
| HR ACTUAL EM (LB/HR) | .000 UNKNOWN        |

|                         |           |
|-------------------------|-----------|
| COMPLIANCE (3=YES;1=NO) | 3 UNKNOWN |
| REV APEN?               | NO        |

## COMMENTS:

1. FILE NAME: 3\_3\_87
2. COAL CRUSHING WILL BE LESS THAN INDICATED SCHEDULE.

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HR  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.255 HRS  
 EIS: 0.50 HR

EIS POINT ID: 4  
 POINT NAME: QUARRIES  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: GRANDFATHERED  
 CONTROL EQUIP: NONE

ROCK BLASTED: 1,400,248 TONS  
 24HRS/7DAY/47WKS/YR=7896HR/YR

|             |         |            |     |    |     |     |      |
|-------------|---------|------------|-----|----|-----|-----|------|
| POLLUTANT   | PART    | OPACITY    | VOC | CO | NOX | SOX | LEAD |
| REGULATION  | NO SPEC | 1,III,2,d  |     |    |     |     |      |
| DESIGN RATE |         |            |     |    |     |     |      |
| HR ALLOW EM |         | 20% and    |     |    |     |     |      |
|             |         | NO OFFPROP |     |    |     |     |      |

|                     |             |      |
|---------------------|-------------|------|
| YR THRUPUT (TONS)   | 1400248     |      |
| REFERENCE           | 3-05-020-09 | INSP |
| EM FACTOR (LB/TON)  | 0.16        |      |
| CONT FACTOR         | 0           |      |
| YR ACTUAL EM (TONS) | 112.020     |      |

|              |         |
|--------------|---------|
| HR ACTUAL EM | UNKNOWN |
|--------------|---------|

|            |    |         |
|------------|----|---------|
| COMPLIANCE | NA | UNKNOWN |
| REV APEN?  |    | NO      |

COMMENTS:  
 1. FILE NAME: 3\_4\_87

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: 05  
 POINT NAME: SITE FUGITIVE DUST  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: NONE  
 CONTROL: FREQ WATER  
 36,000 VMT  
 10HRS/4DAY/47WKS/YR=2000HR/YR

| POLLUTANT   | PART    | OPACITY     | VOC | CO | NOX | SOX | LEAD |
|-------------|---------|-------------|-----|----|-----|-----|------|
| REGULATION  | NO SPEC | 1,III,D,2,e |     |    |     |     |      |
| DESIGN RATE |         |             |     |    |     |     |      |
| HR ALLOW EM |         | NO OFFPROP  |     |    |     |     |      |
| YR ALLOW EM |         |             |     |    |     |     |      |

|                  |              |      |
|------------------|--------------|------|
| YR THRUPUT (VMT) | 36000        |      |
| REFERENCE        | APCD FD MEMO | INSP |
|                  | SECT VI      |      |
|                  | 11.2.1-2.    |      |

|                    |        |
|--------------------|--------|
| EM FACTOR (LB/VMT) |        |
| k (PART SIZE)      | 0.8    |
| s (SILT %)         | 3      |
| S (SPEED)          | 25     |
| W (TRUCK WT)       | 100    |
| w (# WHEELS)       | 6      |
| p (# WET DAYS)     | 82     |
|                    | 10.871 |
| CONT FACTOR        | 0.5    |
| YR ACTUAL EM (TON) | 97.836 |

|           |            |
|-----------|------------|
| HR ACT EM | NO OFFPROP |
|-----------|------------|

|            |    |     |
|------------|----|-----|
| COMPLIANCE | NA | YES |
| REV APEN?  |    |     |

## COMMENTS:

1. FILE NAME: 3\_5\_87
2.  $VMT = (3 \text{ TRUCKS})(30 \text{ TRIPS/TRUCK/DAY})(2 \text{ MI/TRIP})(200 \text{ DAYS}) = VMT$
3. CONTROL EFFICIENCY MAY BE >50%.

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HR  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: 06  
 POINT NAME: LIMESTONE DRYER  
 (AND RAW MTLs GRINDING)

PERMIT: C-12,444(1) MOD  
 CONTROL EQUIP: BAGHOUSE

NSPS REG?: YES: 6,B,IV  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:

CEMENT PRODUCED: 352,161 TONS  
 GAS BURNED: 41,885,000 SCF  
 #20IL BURNED: 0  
 COAL BURNED: 2,218 TONS  
 4062.0 HR/YR

OP'NAL SCHED:

| POLLUTANT                        | PART        | OPACITY  | VOC     | CO          | NOX         | SOX         | LEAD    |
|----------------------------------|-------------|----------|---------|-------------|-------------|-------------|---------|
| REGULATION                       | C-12,444(1) | 1,II,A,1 | NO SPEC | C-12,444(1) | C-12,444(1) | C-12,444(1) | NO SPEC |
| DESIGN RATE (TON/HR)             | 150         |          |         |             |             |             |         |
| YR LIMESTONE LIMIT (TON)         | 717000      |          |         |             |             |             |         |
| YR COAL LIMIT (TON)              | 7446        |          |         |             |             |             |         |
| YR NAT GAS LIMIT (10E6 SCF)      | 50          |          |         |             |             |             |         |
| HR ALLOW EM (LB/HR)              | 13.9        | 20%      |         | 0.9         | 23.3        | 23.9        |         |
| YR ALLOW EM (TON/YR)             | 33.2        |          |         | 2.23        | 55.8        | 57.3        |         |
| ALLOW EM (LB/10E6 BTU)           |             |          |         |             |             | 1.2         |         |
| YR THRUPUT LIMESTONE (TON)       | 520452      |          |         |             |             |             |         |
| YR THRUPUT (TONS CEMENT PROD)    | 352161      |          |         |             |             |             |         |
| REFERENCE                        | 3-05-006-13 | INSP     |         |             |             |             |         |
| EM FACTOR (LB/TON)               | 64          |          |         |             |             |             |         |
| CONT FACTOR                      | 0.999       |          |         |             |             |             |         |
| YR ACTUAL EM (TON)               | 11.269      |          |         |             |             |             |         |
| YR THRUPUT (10E6 SCF GAS BURNED) | 41.885      |          | 41.885  | 41.885      | 41.885      | 41.885      |         |
| REFERENCE                        | 1-02-006-01 |          | SAME    | SAME        | SAME        | SAME        |         |
| EM FACTOR (LB/10E6 SCF)          | 3           |          | 1.4     | 40          | 550         | 0.6         |         |
| CONT FACTOR                      | 0.999       |          | 0       | 0           | 0           | 0           |         |
| YR ACTUAL EM (TON)               | .000        |          | 0.029   | 0.838       | 11.518      | 0.013       |         |
| BTU VALUE (BTU/SCF)              |             |          |         |             |             | 991         |         |
| YR HEAT INPUT (10E6 BTU/YR)      |             |          |         |             |             | 41508.035   |         |
| YR THRUPUT (TON COAL BURNED)     | 2218        |          | 2218    | 2218        | 2218        | 2218        |         |
| REFERENCE                        | 3-90-002-01 |          | SAME    | SAME        | SAME        | SAME        |         |
| EM FACTOR (LB/TON)               | 0           |          | 0       | 0           | 0           | 26          |         |
| % SULFUR                         |             |          |         |             |             | 0.4         |         |
| CONT FACTOR                      | 0.999       |          | 0       | 0           | 0           | 0           |         |
| YR ACTUAL EM (TON)               | 0.000       |          | 0.000   | 0.000       | 0.000       | 11.534      |         |

|                             |               |       |       |        |        |    |
|-----------------------------|---------------|-------|-------|--------|--------|----|
| BTU VALUE (BTU/TON)         |               |       |       |        | 24     |    |
| YR HEAT INPUT (10E6 BTU/YR) |               |       |       |        | 53232  |    |
| YR ACTUAL EM TOTAL (TONS)   | 11.269        | 0.029 | 0.838 | 11.518 | 11.546 |    |
| ACT EM (LB/10E6 BTU)        |               |       |       |        | 0.434  |    |
| TIME FACTOR (HR/YR)         | 4062          |       |       |        |        |    |
| HR ACTUAL EM (LB/HR)        | 5.549         | 0%    |       |        |        |    |
| COMPLIANCE (3=YES;1=NO)     |               |       |       |        |        |    |
| YR LIMESTONE LIMIT          | 3             |       |       |        |        |    |
| YR COAL LIMIT               | 3             |       |       |        |        |    |
| YR NAT GAS LIMIT            | 3             |       |       |        |        |    |
| HR                          | 3             | YES   | 3     | 3      | 3      | NA |
| YR                          | 3             |       | 3     | 3      | 3      |    |
| LB/BTU                      |               |       |       |        | 3      |    |
| REV APEN? (3=YES;1=NO)      | YES           |       |       |        |        |    |
| REFERENCE                   | 3,II,B,2,b,ii |       |       |        |        |    |

COMMENTS:

1. FILE NAME: 3\_6\_87
2. EM FACTORS FOR PART, VOC, CO, AND NOX FOR FUELS  
ARE NOT PROVIDED IN SCC CODE BOOK.

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9\_20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HR  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: UNASSIGNED, 07  
 POINT NAME: HOMOGENIZING SILOS  
 MSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: P-10,284  
 CONTROL EQUIP: BAGHOUSE

559,886 TON MTL  
 24HRS/7DAY/47WKS/YR=7896HR/YR

| POLLUTANT            | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|----------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION           | 1,III,C,1,b | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TON/HR) | 120         |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)  | 37.236      | 20%      |     |    |     |     |      |

|             |           |      |
|-------------|-----------|------|
| YR THRUPUT  |           |      |
| REFERENCE   |           | INSP |
| EM FACTOR   | NONE ID'D |      |
| CONT FACTOR |           |      |
| YR ACT EM   |           |      |
| TIME FACTOR |           |      |
| HR ACT EM   |           | 0%   |

COMPLIANCE YES  
 REV APEN?

COMMENTS:  
 1. FILE NAME: 3\_7\_87

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HR  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: UNASSIGNED, 08  
 POINT NAME: A-FRAME AND  
 CLINKER STORAGE  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: A-FRAME: P-10,226; P-10,292  
 OUTSIDE: 84B0369F  
 CONTROL EQUIP: DUST COLLECTORS  
 A-FRAME: 280,000 TONS  
 OUTSIDE: EST 30,000 TONS  
 24HRS/7DAY/52WKS/YR=8760HR/YR

| POLLUTANT<br>REGULATION | PART<br>84B0369F | OPACITY<br>1,II,A,1<br>1,III,D,2,c | VOC | CO | NOX | SOX | LEAD |
|-------------------------|------------------|------------------------------------|-----|----|-----|-----|------|
|-------------------------|------------------|------------------------------------|-----|----|-----|-----|------|

|                              |        |                      |  |  |  |  |  |
|------------------------------|--------|----------------------|--|--|--|--|--|
| LOADOUT DESIGN RATE (TON/HR) | 100    |                      |  |  |  |  |  |
| MAX OUTSIDE STORAGE (TON)    | 120000 |                      |  |  |  |  |  |
| HR ALLOW EM (LB/HR)          | 12.33  | 20% & NO<br>OFF PROP |  |  |  |  |  |
| YR ALLOW EM (TON/YR)         | 10.2   |                      |  |  |  |  |  |

|                       |       |                       |  |  |  |  |  |
|-----------------------|-------|-----------------------|--|--|--|--|--|
| OUTSIDE PILES         |       |                       |  |  |  |  |  |
| YR THRUPUT (TON)      | 30000 |                       |  |  |  |  |  |
| REFERENCE             |       | INSP                  |  |  |  |  |  |
| EM FACTOR (LB/TON)    | 0.1   |                       |  |  |  |  |  |
| CONT FACTOR           | 0     |                       |  |  |  |  |  |
| YR ACTUAL EM (TON/YR) | 1.5   |                       |  |  |  |  |  |
| TIME FACTOR (LB/YR)   | 8760  |                       |  |  |  |  |  |
| HR ACTUAL EM (LB/HR)  | 0.342 | (20% & NO<br>OFF PROP |  |  |  |  |  |

|                         |    |     |  |  |  |  |  |
|-------------------------|----|-----|--|--|--|--|--|
| COMPLIANCE (3=YES;1=NO) |    |     |  |  |  |  |  |
| OUTSIDE STORAGE LIMIT   | 3  |     |  |  |  |  |  |
| HR                      | 3  |     |  |  |  |  |  |
| YR                      | 3  |     |  |  |  |  |  |
|                         | NO | YES |  |  |  |  |  |

REV APEN?

## COMMENTS:

1. FILE NAME: 3\_8\_87
2. TRANSFER OF MTL BY FRONTLOADER INTO OUTSIDE HOPPER NOT OBSERVED



# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HR  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: 09  
 POINT NAME: KILN STACK  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:

PERMIT: C-12,444(2)  
 CONTROL EQUIP: BAGHOUSE

KILN FEED: 493,292 TONS  
 CEMENT PRODUCED: 352,161 TONS  
 COAL BURNED: 46,481 TONS (0.5% SULFUR; 8% ASH; 14,675 BTU/LB)  
 #2OIL BURNED: 2,125,467 GAL (0.36% SULFUR); 133,000 BTU/GAL  
 GAS BURNED: 92,223,000 SCF (1075 BTU/SCF)  
 5061.5 HR/YR

## OPERATIONAL SCHEDULE:

| POLLUTANT<br>REGULATION         | PART<br>C-12,444(2) | OPACITY<br>1,II,A,1 | VOC<br>NO SPEC | CO<br>NO SPEC | NOX<br>NO SPEC | SOX<br>C-12,444(2) | LEAD        |
|---------------------------------|---------------------|---------------------|----------------|---------------|----------------|--------------------|-------------|
| DESIGN RATE (TON/HR)            | 102                 |                     |                |               |                |                    |             |
| HR ALLOW EM (LB/HR)             | 27                  | 20%                 |                |               |                |                    |             |
| YR ALLOW EM (TON/YR)            | 102                 |                     |                |               |                | 851                |             |
| ALLOW EM (LB/TON FEED TO KILN)  | 0.3                 |                     |                |               |                |                    |             |
| ALLOW EM (LB/10E6 BTU)          |                     |                     |                |               |                | 1.2                |             |
| YR THRUPUT (TONS OF CEM PROD)   | 352161              |                     |                |               | 352161         | 352161             | 352161      |
| REFERENCE                       | 3-05-006-06         | INSP                |                |               | 3-05-006-06    | 3-05-006-06        | 3-05-006-06 |
| EM FACTOR (LB/TON)              | 245                 |                     | 0              | 0             | 2.6            | 10.2               | 0.12        |
| CONT FACTOR                     | 0.999               |                     |                |               | 0              | 0.75               | 0.999       |
| YR ACTUAL EM (TONS)             | 43.140              |                     |                |               | 457.809        | 449.005            | 0.021       |
| TIME FACTOR (HRS/YR)            | 5061.5              |                     |                |               |                |                    |             |
| HR ACTUAL EM (LB/HR)            | 17.046              | 0%                  |                |               |                |                    |             |
| KILN FEED (TON)                 | 493292              |                     |                |               |                |                    |             |
| ACTUAL EM (LB/TON FEED TO KILN) | 0.175               |                     |                |               |                |                    |             |
| YR THRUPUT (TONS COAL BURNED)   |                     |                     |                |               |                | 46481              |             |
| REFERENCE                       |                     |                     |                |               |                | 3-90-002-01        |             |
| EM FACTOR (LB/TON)              |                     |                     |                |               |                | 26                 |             |
| %S                              |                     |                     |                |               |                | 0.5                |             |
| CONT FACTOR                     |                     |                     |                |               |                | 0.75               |             |
| YR ACTUAL EM (TON)              |                     |                     |                |               |                | 75.532             |             |
| BTU VALUE (10E6 BTU/TON)        |                     |                     |                |               |                | 29.35              |             |
| YR HEAT INPUT (10E6 BTU)        |                     |                     |                |               |                | 1364217.35         |             |
| YR THRUPUT (10E3 GAL BURNED)    |                     |                     |                |               |                | 2125.467           |             |
| REFERENCE                       |                     |                     |                |               |                | 3-90-004-02        |             |
| EM FACTOR (LB/10E3 GAL)         |                     |                     |                |               |                | 108                |             |

|                                |               |         |               |
|--------------------------------|---------------|---------|---------------|
| ZS                             |               |         | 0.36          |
| CONT FACTOR                    |               |         | 0.75          |
| YR ACTUAL EM (TON)             |               |         | 10.330        |
| BTU VALUE (10E6 BTU/GAL)       |               |         | 0.133         |
| YR HEAT INPUT (10E6 BTU)       |               |         | 282687.111    |
| YR THRUPUT (10E6 SCF BURNED)   |               |         | 92.223        |
| BTU VALUE (BTU/SCF)            |               |         | 1075          |
| YR HEAT INPUT (10E6 BTU)       |               |         | 99139.725     |
| YR ACT EM TOTAL (TONS)         | 43.140        | 457.809 | 534.867       |
| YR HEAT INPUT TOTAL (10E6 BTU) |               |         | 1746044.186   |
| YR ACTUAL EM (LB/10E6 BTU)     |               |         | 0.613         |
| COMPLIANCE (3=YES;1=NO) YR     | 3             | YES     | 3             |
| HR                             | 3             |         | NA            |
| FEED                           | 3             |         | NA            |
| BTU                            | NA            |         | 3             |
| REV APEN? (3=YES;1=NO)         | 1             |         | 3             |
| REFERENCE                      | 3,II,8,2,b,ii |         | 3,II,8,2,b,iv |
| COMMENTS:                      |               |         |               |
| 1. FILE NAME: 3_9_87           |               |         |               |

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.255 HRS  
 EIS: 0.50 HRS

EIS POINT ID: 10  
 POINT NAME: FINISHED PRODUCTS  
 SILOS

PERMIT: GRANDFATHERED  
 CONTROL EQUIP: BAGHOUSE

NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:

TOTAL CEMENT PRODUCED: 352,161 TONS

OP'NAL SCHED:

24HRS/7DAY/47WKS/YR=7896HR/YR

| POLLUTANT               | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|-------------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION              | 1,III,C,1,b | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TON/HR)    | 85          |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)     | 35.237      | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)       | 352161      |          |     |    |     |     |      |
| REFERENCE               | 3-05-006-19 | INSP     |     |    |     |     |      |
| EM FACTOR (LB/TON)      | 0.24        |          |     |    |     |     |      |
| CONT FACTOR             | 0.999       |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)     | 0.042       |          |     |    |     |     |      |
| TIME FACTOR (HR/YR)     | 7896        |          |     |    |     |     |      |
| HR ACTUAL EM (LB/HR)    | 0.011       | 0%       |     |    |     |     |      |
| COMPLIANCE (3=YES;1=NO) | 3           | YES      |     |    |     |     |      |
| REV APEN?               | NO          |          |     |    |     |     |      |

COMMENTS:

1. FILE NAME: 3\_10\_87

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20-88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.50 HRS

EIS POINT ID: 11  
 POINT NAME: MILL BUILDING  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: P-10,078  
 CONTROL EQUIP: BAGHOUSE

TOTAL CEMENT PRODUCED: 352,161 TONS  
 24HRS/7DAY/47WKS/YR=7896HR/YR

| POLLUTANT               | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|-------------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION              | 1,III,C,1,b | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TON/HR)    | 85          |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)     | 35.237      | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)       | 352161      |          |     |    |     |     |      |
| REFERENCE               | 3-05-006-17 | INSP     |     |    |     |     |      |
| EM FACTOR (LB/TON)      | 96          |          |     |    |     |     |      |
| CONT FACTOR             | 0.999       |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)     | 16.904      |          |     |    |     |     |      |
| TIME FACTOR (HR/YR)     | 7896        |          |     |    |     |     |      |
| HR ACTUAL EM (LB/HR)    | 4.282       | 0%       |     |    |     |     |      |
| COMPLIANCE (3=YES;1=NO) | 3           | YES      |     |    |     |     |      |
| REV APEN?               | NO          |          |     |    |     |     |      |

COMMENTS:  
 1. FILE NAME: 3\_11\_87

# AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3  
 SOURCE NAME: SW PORTLAND CEMENT CO  
 LOCATION: LYONS  
 COUNTY: BLDR 0220  
 CONTACT: RANDY WILEY  
 TELEPHONE: 534-4206  
 DATE: 9-20\_88  
 TIME: 9:00-1:00

INSPECT TYPE: ANNUAL  
 ACTION TYPE: 12  
 INSPECTOR: G. MATHEWS  
 HRS REQUIRED: 12.75 HRS  
 PREPARATION: 1.00 HRS  
 TRAVEL: 1.00 HRS  
 INSPECTION: 4.00 HRS  
 WRITE UP: 6.25 HRS  
 EIS: 0.500 HRS

EIS POINT ID: UNASSIGNED,13  
 POINT NAME: SILICA SILOS  
 NSPS REG?: NO  
 NESHAPS REG?: NO  
 PSD REG?: NO  
 1987 ANNUAL THRUPUT:  
 OP'NAL SCHED:

PERMIT: GRANDFATHERED  
 CONTROL EQUIP: BAGHOUSE

38,434 TONS  
 24HRS/7DAY/47WKS/YR=7896HR/YR

| POLLUTANT             | PART        | OPACITY  | VOC | CO | NOX | SOX | LEAD |
|-----------------------|-------------|----------|-----|----|-----|-----|------|
| REGULATION            | 1,III,C,1,a | 1,II,A,1 |     |    |     |     |      |
| DESIGN RATE (TONS/HR) | 4           |          |     |    |     |     |      |
| HR ALLOW EM (LB/HR)   | 8.480       | 20%      |     |    |     |     |      |
| YR THRUPUT (TONS)     | 38434       |          |     |    |     |     |      |
| REFERENCE             |             | INSP     |     |    |     |     |      |
| EM FACTOR             | NONE ID'D   |          |     |    |     |     |      |
| CONT FACTOR           |             |          |     |    |     |     |      |
| YR ACTUAL EM (TONS)   |             |          |     |    |     |     |      |
| TIME FACTOR (HR/YR)   |             |          |     |    |     |     |      |
| HR ACTUAL EM (LB/HR)  |             | 0-10%    |     |    |     |     |      |
| COMPLIANCE            | UNKNOWN     | YES      |     |    |     |     |      |
| REV APEN?             |             | NO       |     |    |     |     |      |

COMMENTS  
 1. FILE NAME: 3\_12\_87

SW PORTLAND CEMENT CO., SOURCE 3  
1987 EMISSION INVENTORY

| EIS PT ID       | NAME                         | POLLUTANT(TONS) |         |       |       |         |      |
|-----------------|------------------------------|-----------------|---------|-------|-------|---------|------|
|                 |                              | PART            | SOX     | VOC   | CO    | NOX     | LEAD |
| 01              | CLINKER COOLER               | 1.620           |         |       |       |         |      |
| 02 (UNASSIGNED) | RAW MTL CRUSHERS             | 0.644           |         |       |       |         |      |
| 03 (UNASSIGNED) | COAL HANDLING                | 0.000           |         |       |       |         |      |
| 04              | QUARRIES                     | 112.020         |         |       |       |         |      |
| 05              | SITE FUG DUST                | 97.836          |         |       |       |         |      |
| 06              | LIMESTONE DRYER              | 11.269          | 11.546  | 0.029 | 0.838 | 11.518  |      |
| 07 (UNASSIGNED) | HOMOGENIZING SILOS           |                 |         |       |       |         |      |
| 08 (UNASSIGNED) | A-FRAME & CLINKER<br>STORAGE | 1.500           |         |       |       |         |      |
| 09              | KILN STACK                   | 43.140          | 534.867 |       |       | 457.809 |      |
| 10              | FIN PRODUCTS SILOS           | 0.042           |         |       |       |         |      |
| 11              | MILL BUILDING                | 16.904          |         |       |       |         |      |
| 12 (UNASSIGNED) | SILICA SILOS                 |                 |         |       |       |         |      |
|                 | TOTAL                        | 284.975         | 546.413 | 0.029 | 0.838 | 469.327 |      |

COMMENTS:

1. FILE NAME: 3\_87

# OPERATIONS REPORT - LYONS PLA

MONTHLY

YEAR-TO-DATE

|            |              |         |             |                 |             |             |
|------------|--------------|---------|-------------|-----------------|-------------|-------------|
| I.R. Drill | Ft: 5,471    | Hrs: 98 | Ft/Hr: 55.8 | Ft: 68,620      | Hrs: 1074.0 | Ft/Hr: 63.9 |
| Shipping   | Tons: 86,471 |         |             | Tons: 1,265,437 |             |             |

|              | TONS   | HOURS | TONS/HR. | % OPER. | TONS    | HOURS  | TONS/HR. | % OPER. |
|--------------|--------|-------|----------|---------|---------|--------|----------|---------|
| SHING:       |        | 744   |          |         |         | 8760   |          |         |
| Cal.R#1      | 25,609 | 194.5 | 131.7    | 26.2    | 293,323 | 2098.0 | 139.8    | 23.9    |
| Hi.Cem.R#3   | 13,819 | 105.0 | 131.6    | 14.1    | 135,158 | 974.0  | 138.8    | 11.1    |
| Lo Cem.R#3-4 | 0      |       |          |         | 17,740  | 127.0  | 139.7    | 1.5     |
| Weathered    | 8,472  | 73.5  | 115.3    | 9.9     | 75,231  | 596.5  | 126.1    | 6.8     |
| R. #4        | 0      |       |          |         | 0       |        |          |         |
| Silica       | 826    | 6.0   | 137.7    | 0.8     | 9,867   | 71.0   | 139.0    | 0.8     |
| Gypsum       | 1,281  | 10.5  | 122.0    | 1.4     | 20,970  | 138.0  | 151.9    | 1.6     |
| Iron         | 761    | 5.5   | 138.4    | 0.7     | 7,597   | 57.5   | 132.1    | 0.7     |
| TOTALS       | 50,768 | 395.0 | 128.5    | 53.1    | 559,886 | 4062.0 | 137.8    | 46.4    |

|           |        |       |       |      |         |        |       |      |
|-----------|--------|-------|-------|------|---------|--------|-------|------|
| RAW GRIND | 44,463 | 365.0 | 121.8 | 49.1 | 493,292 | 3896.5 | 126.6 | 44.5 |
| KILN      | 24,415 | 492.5 | 49.6  | 66.2 | 280,607 | 5061.5 | 55.4  | 57.8 |

|              |        |       |      |      |         |        |      |      |
|--------------|--------|-------|------|------|---------|--------|------|------|
| FINISH MILL: |        |       |      |      |         |        |      |      |
| I            | 1,096  | 13.0  | 84.3 | 1.7  | 34,057  | 380.0  | 89.6 | 4.3  |
| I LA         | 0      |       |      |      | 0       |        |      |      |
| II           | 7,189  | 84.5  | 85.1 | 11.4 | 97,766  | 1159.0 | 84.5 | 13.2 |
| II LA        | 7,988  | 97.0  | 82.4 | 13.1 | 160,977 | 1951.5 | 82.5 | 22.3 |
| II MOD       | 1,788  | 21.0  | 85.1 | 2.8  | 30,794  | 350.0  | 88.0 | 4.0  |
| III LA       | 0      |       |      |      | 8,675   | 187.5  | 46.1 | 2.2  |
| PRODUCT      | 653    | 12.0  | 54.4 | 1.6  | 16,837  | 319.5  | 52.7 | 3.7  |
| ON           | 0      |       |      |      | 3,055   | 55.0   | 55.5 | 0.6  |
| TOTALS       | 18,714 | 227.5 | 82.3 | 30.6 | 352,161 | 4402.5 | 80.0 | 50.3 |

iv. Prod: .85x 24,415 + .15x 18,714 = 23,560 Tons      .85x 280,607 + .15x 352,161 = 291,340 Tons

|                                      |                  |         |                    |           |                    |           |                    |           |
|--------------------------------------|------------------|---------|--------------------|-----------|--------------------|-----------|--------------------|-----------|
| POWER:                               | KWH: 3,564,000   |         | COST/KWH: 0.0684   |           | KWH: 57,951,000    |           | COST/KWH: 0.0508   |           |
|                                      | COST: 243,813.00 |         | COST/EQ.TON: 10.35 |           | COST: 2,942,297.91 |           | COST/EQ.TON: 10.10 |           |
| FUEL:<br>DRYER<br>KILN<br><br>TOTALS | 1014             | 131,000 | 12,400             |           | 1008               | 131,000   | 11,902             |           |
|                                      | GAS MCF          | OIL/GAL | COAL/TONS          | BTU/TON   | GAS MCF            | OIL/GAL   | COAL/TONS          | BTU/TON   |
|                                      | 6,125            | 0       | 165                | 202,938   | 41,885             | 0         | 2,218              | 169,708   |
|                                      | 4,766            | 298,624 | 3,517              | 5,372,684 | 92,223             | 2,125,467 | 46,481             | 5,266,550 |
|                                      | 10,891           | 298,624 | 3,682              | 5,794,668 | 134,108            | 2,125,467 | 48,699             | 5,605,163 |

|           |          |              |                     |             |                    |          |              |                     |             |                    |
|-----------|----------|--------------|---------------------|-------------|--------------------|----------|--------------|---------------------|-------------|--------------------|
| MANHOURS: | No. Emp. | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hrs/ Equiv. T. | No. Emp. | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hrs/ Equiv. T. |
| Hourly    | 69       | 8566         | 2307                | 10873       | .46                |          | 145,676      | 23,678              | 169354      | .58                |
| Staff     | 35       | -            | -                   | 6428        | .27                |          | -            | -                   | 78828       | .27                |
| TOTALS    | 104      | -            | -                   | 17301       | .73                |          | -            | -                   | 248182      | .85                |

|        |                           |
|--------|---------------------------|
| SALES: | Overtime Hours: 658.5     |
|        | MONTHLY      YEAR-TO-DATE |
| I      | 908      37,411           |
| I LA   | 0      0                  |
| II     | 3,289      116,258        |
| II LA  | 7,604      147,987        |
| "G"    | 1,446      7,004          |
| II MOD | 1,390      27,069         |
| DUCT   | 744      17,384           |
| ONS    | 93      3,708             |
| or     | 0      72                 |
| TOTALS | 15,474      356,893       |

|                                    |         |              |
|------------------------------------|---------|--------------|
| SAFETY:                            | MONTHLY | YEAR-TO-DATE |
| First Aid Cases                    | 4       | 92           |
| Doctor Cases                       | 0       | 11           |
| Lost Time Accidents                | 0       | 4            |
| Days since last lost time accident | 85      | 85           |

*E. D. Cook*  
Manager of Operations

MONTH: DECEMBER 1987

OPERATIONS REPORT - LYONS PLA

MONTHLY

YEAR-TO-DATE

|                                        |      |       |          |         |                                                                           |              |                     |             |                    |                                                                            |              |                     |             |               |
|----------------------------------------|------|-------|----------|---------|---------------------------------------------------------------------------|--------------|---------------------|-------------|--------------------|----------------------------------------------------------------------------|--------------|---------------------|-------------|---------------|
| I.R. Drill<br>ipping                   |      |       |          |         | Ft: 10,129   Hrs: 118   Ft/Hr: 85.8<br>Tons: 99,477                       |              |                     |             |                    | Ft: 16,885   Hrs: 241.5   Ft/Hr: 69.9<br>Tons: 296,030                     |              |                     |             |               |
| ISHING:                                |      |       |          |         | 744                                                                       |              |                     |             |                    | 2160                                                                       |              |                     |             |               |
|                                        | TONS | HOURS | TONS/HR. | % OPER. |                                                                           |              |                     |             |                    | TONS                                                                       | HOURS        | TONS/HR.            | % OPER.     |               |
| Cal.R#1                                | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 19,748                                                                     | 150.5        | 131.2               | 7.0         |               |
| Hi.Cem.R#3                             | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 13,928                                                                     | 106.5        | 130.8               | 4.9         |               |
| Lo.Cem.R#3-4                           | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 2,178                                                                      | 16.5         | 132.0               | 0.8         |               |
| Weathered                              | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 7,106                                                                      | 67.5         | 105.3               | 3.1         |               |
| R. #4                                  | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 0                                                                          |              |                     |             |               |
| Silica                                 | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 695                                                                        | 6.0          | 115.8               | 0.3         |               |
| Gypsum                                 | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 889                                                                        | 8.5          | 104.6               | 0.4         |               |
| Iron                                   | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 364                                                                        | 3.0          | 121.3               | 0.1         |               |
| TOTALS                                 | 0    | 0     | 0        | 0       |                                                                           |              |                     |             |                    | 44,908                                                                     | 358.5        | 125.3               | 16.6        |               |
| RAW GRIND                              |      |       |          |         | 0                                                                         | 0            | 0                   | 0           |                    | 41,358                                                                     | 355.5        | 116.3               | 16.5        |               |
| KILN                                   |      |       |          |         | 0                                                                         | 0            | 0                   | 0           |                    | 25,535                                                                     | 472.0        | 54.1                | 21.9        |               |
| FINISH MILL:                           |      |       |          |         |                                                                           |              |                     |             |                    |                                                                            |              |                     |             |               |
| I                                      | 730  | 9.0   | 81.1     | 1.2     |                                                                           |              |                     |             |                    | 6,550                                                                      | 72.0         | 91.0                | 3.3         |               |
| I LA                                   | 0    |       |          |         |                                                                           |              |                     |             |                    | 0                                                                          |              |                     |             |               |
| II                                     | 4165 | 67.5  | 61.7     | 9.1     |                                                                           |              |                     |             |                    | 16,135                                                                     | 200.0        | 80.7                | 9.3         |               |
| II LA                                  | 480  | 6.0   | 80.0     | 0.8     |                                                                           |              |                     |             |                    | 11,148                                                                     | 131.5        | 84.8                | 6.1         |               |
| II MOD                                 | 0    |       |          |         |                                                                           |              |                     |             |                    | 3,397                                                                      | 43.0         | 79.0                | 2.0         |               |
| Class G                                | 0    |       |          |         |                                                                           |              |                     |             |                    | 0                                                                          |              |                     |             |               |
| DUCT                                   | 1271 | 25.5  | 49.8     | 3.4     |                                                                           |              |                     |             |                    | 3,485                                                                      | 70.0         | 49.8                | 3.2         |               |
| ON                                     | 0    |       |          |         |                                                                           |              |                     |             |                    | 765                                                                        | 13.0         | 58.8                | 0.6         |               |
| TOTALS                                 | 6646 | 108.0 | 61.5     | 14.5    |                                                                           |              |                     |             |                    | 41,480                                                                     | 529.5        | 78.3                | 24.5        |               |
| iv. Prod: .85x 0 +.15x 6646 = 997 Tons |      |       |          |         | .85x 25,535 +.15x 41,480 = 27,927 Tons                                    |              |                     |             |                    |                                                                            |              |                     |             |               |
| POWER:                                 |      |       |          |         | KWH: 918,000   COST/KWH: 0.1103<br>COST: 101,340.73   COST/EQ.TON: 101.65 |              |                     |             |                    | KWH: 8,064,000   COST/KWH: 0.0577<br>COST: 465,794.88   COST/EQ.TON: 16.68 |              |                     |             |               |
| FUEL:                                  |      |       |          |         | GAS MCF OIL/GAL COAL/TONS BTU/TON                                         |              |                     |             |                    | 992,000 135,000 10,480<br>GAS MCF OIL/GAL COAL/TONS BTU/TON                |              |                     |             |               |
| DRYER                                  |      |       |          |         | 0                                                                         | 0            | 0                   | 0           |                    | 1995                                                                       | -            | 280                 | 221,427     |               |
| KILN                                   |      |       |          |         | 0                                                                         | 0            | 0                   | 0           |                    | 2373                                                                       | 208,204      | 4825                | 5,153,458   |               |
| TOTALS                                 |      |       |          |         | 0                                                                         | 0            | 0                   | 0           |                    | 4368                                                                       | 208,204      | 5205                | 5,542,878   |               |
| MANHOURS:                              |      |       |          |         | No. Emp.                                                                  | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hrs/ Equiv. T. | No. Emp.                                                                   | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hr Equiv. |
| Hourly                                 |      |       |          |         | 85                                                                        | 11,099       | 619                 | 11718       | 11.75              |                                                                            | 34,501       | 3497                | 37998       | 1.36          |
| Staff                                  |      |       |          |         | 38                                                                        | -            | -                   | 6589        | 6.61               |                                                                            | -            | -                   | 19941       | .71           |
| TOTALS                                 |      |       |          |         | 123                                                                       | -            | -                   | 18307       | 18.36              |                                                                            | -            | -                   | 57939       | 2.07          |
| SALES:                                 |      |       |          |         | Overtime Hrs. 8.0<br>MONTHLY YEAR-TO-DATE                                 |              |                     |             |                    | SAFETY:                                                                    |              |                     |             |               |
| I                                      |      |       |          |         | 2,588                                                                     | 7,886        |                     |             |                    | First Aid Cases                                                            |              |                     |             |               |
| I LA                                   |      |       |          |         | 0                                                                         | 0            |                     |             |                    | Doctor Cases                                                               |              |                     |             |               |
| II                                     |      |       |          |         | 6,627                                                                     | 16,921       |                     |             |                    | Lost Time Accidents                                                        |              |                     |             |               |
| II LA                                  |      |       |          |         | 6,692                                                                     | 16,258       |                     |             |                    | Days since last lost time accident                                         |              |                     |             |               |
| "G"                                    |      |       |          |         | 24                                                                        | 80           |                     |             |                    |                                                                            |              |                     |             |               |
| MOD                                    |      |       |          |         | 1,357                                                                     | 4,405        |                     |             |                    |                                                                            |              |                     |             |               |
| DUCT                                   |      |       |          |         | 1,288                                                                     | 3,396        |                     |             |                    |                                                                            |              |                     |             |               |
| ONS                                    |      |       |          |         | 219                                                                       | 668          |                     |             |                    |                                                                            |              |                     |             |               |
| er                                     |      |       |          |         | 9                                                                         | 31           |                     |             |                    |                                                                            |              |                     |             |               |
| TOTALS                                 |      |       |          |         | 18,804                                                                    | 49,645       |                     |             |                    |                                                                            |              |                     |             |               |
|                                        |      |       |          |         |                                                                           |              |                     |             |                    | E. D. Cook, Manager of Operations                                          |              |                     |             |               |
|                                        |      |       |          |         |                                                                           |              |                     |             |                    | MONTH: MARCH 1987                                                          |              |                     |             |               |



OPERATIONS REPORT - LYONS PLA

MONTHLY

YEAR-TO-DATE

|            |               |            |             |               |            |             |
|------------|---------------|------------|-------------|---------------|------------|-------------|
| I.R. Drill | Ft: 7,976     | Hrs: 114.5 | Ft/Hr: 69.7 | Ft: 38,404    | Hrs: 540.5 | Ft/Hr: 71.1 |
| ipping     | Tons: 158,819 |            |             | Tons: 680,584 |            |             |

| SHING:        | TONS   | 720 HOURS | TONS/HR. | % OPER. | TONS    | 4344 HOURS | TONS/HR. | % OPER. |
|---------------|--------|-----------|----------|---------|---------|------------|----------|---------|
| ... Cal. R#1  | 43,869 | 293.5     | 149.5    | 40.8    | 103,722 | 737.0      | 140.7    | 17.0    |
| Hi. Cem. R#3  | 13,209 | 88.0      | 150.1    | 12.2    | 40,954  | 295.0      | 138.8    | 6.8     |
| Lo Cem. R#3-4 | 0      |           |          |         | 5,509   | 41.0       | 134.4    | 0.9     |
| Weathered     | 8,883  | 68.0      | 130.6    | 9.4     | 22,306  | 189.0      | 118.0    | 4.4     |
| R. #4         | 0      |           |          |         | 0       |            |          |         |
| Silica        | 789    | 6.0       | 131.5    | 0.8     | *2,724  | 21.0       | 129.7    | 0.5     |
| Gypsum        | 2,648  | 21.0      | 126.1    | 2.9     | *9,779  | 54.5       | 179.4    | 1.2     |
| Iron          | 1,181  | 9.0       | 131.2    | 1.3     | *2,498  | 20.0       | 124.9    | 0.5     |
| TOTALS        | 70,579 | 485.5     | 145.4    | 67.4    | 187,492 | 1357.5     | 138.1    | 31.3    |

|              |        |       |       |      |         |        |       |      |
|--------------|--------|-------|-------|------|---------|--------|-------|------|
| RAW GRIND    | 61,484 | 469.0 | 131.1 | 65.1 | 163,780 | 1286.5 | 127.3 | 29.6 |
| KILN         | 34,800 | 616.5 | 56.4  | 85.6 | 93,420  | 1686.0 | 55.4  | 38.8 |
| FINISH MILL: |        |       |       |      |         |        |       |      |
| I            | 6,050  | 62.0  | 97.6  | 8.6  | 18,671  | 207.0  | 90.2  | 4.8  |
| I LA         | 0      |       |       |      | 0       |        |       |      |
| II           | 8,454  | 86.5  | 97.7  | 12.0 | 43,332  | 512.0  | 84.6  | 11.8 |
| II LA        | 15,586 | 179.0 | 87.1  | 24.9 | 51,789  | 626.0  | 82.7  | 14.5 |
| II MOD       | 2,070  | 23.0  | 90.0  | 3.2  | 11,556  | 134.5  | 85.9  | 3.1  |
| III LA       | 1,089  | 24.0  | 45.4  | 3.3  | 2,416   | 54.0   | 44.7  | 1.2  |
| DUCT         | 1,689  | 30.5  | 55.4  | 4.3  | 8,230   | 162.0  | 50.8  | 3.7  |
| ON           | 876    | 14.0  | 62.6  | 1.9  | 1,641   | 27.0   | 60.8  | 0.6  |
| TOTALS       | 35,814 | 419.0 | 85.5  | 58.2 | 137,635 | 1722.5 | 79.9  | 39.7 |

\*All rock used shown as crusher throughput in order to keep books balanced.

iv. Prod:  $.85x 34,800 + .15x 35,814 = 34,952$  Tons

$.85x 93,420 + .15x 137,635 = 100,052$  Tons

|                                      |                  |                   |           |                    |                    |         |           |           |
|--------------------------------------|------------------|-------------------|-----------|--------------------|--------------------|---------|-----------|-----------|
| POWER:                               | KWH: 6,489,000   | COST/KWH: 0.0472  |           | KWH: 21,294,000    | COST/KWH: 0.0546   |         |           |           |
|                                      | COST: 306,556.17 | COST/EQ.TON: 8.77 |           | COST: 1,163,146.62 | COST/EQ.TON: 11.63 |         |           |           |
| FUEL:<br>DRYER<br>KILN<br><br>TOTALS | 1005             | 135,000           | 11,666    | 1011               | 135,000            | 11,473  |           |           |
|                                      | GAS MCF          | OIL/GAL           | COAL/TONS | BTU/TON            | GAS MCF            | OIL/GAL | COAL/TONS | BTU/TON   |
|                                      | 6,496            | -                 | 300       | 197,913            | 17,105             | -       | 723       | 180,718   |
|                                      | 23,969           | 353,616           | 5,185     | 5,540,494          | 53,436             | 887,750 | 14,413    | 5,401,314 |
|                                      | 30,465           | 353,616           | 5,485     | 5,941,888          | 70,541             | 887,750 | 15,136    | 5,764,010 |

|           |          |              |                     |             |                    |          |              |                     |             |                    |
|-----------|----------|--------------|---------------------|-------------|--------------------|----------|--------------|---------------------|-------------|--------------------|
| MANHOURS: | No. Emp. | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hrs/ Equiv. T. | No. Emp. | Hours Worked | Hrs. Pd. Not Worked | Total Hours | Man Hrs/ Equiv. T. |
| Hourly    | 86       | 13,364       | 1,569               | 14933       | .43                |          | 72,947       | 8,320               | 81267       | .81                |
| Staff     | 38       | -            | -                   | 6585        | .19                |          | -            | -                   | 39700       | .40                |
| TOTALS    | 124      | -            | -                   | 21518       | .62                |          | -            | -                   | 120967      | 1.21               |

|        |                      |              |
|--------|----------------------|--------------|
| SALES: | Overtime Hours 321.0 |              |
|        | MONTHLY              | YEAR-TO-DATE |
| I      | 6,248                | 20,284       |
| I LA   | 0                    | 0            |
| II     | 13,442               | 52,671       |
| II LA  | 14,490               | 49,666       |
| "G"    | 0                    | 109          |
| MOD    | 3,535                | 11,737       |
| DUCT   | 2,374                | 8,538        |
| ONS    | 489                  | 1,932        |
| er     | 3                    | 36           |
| TOTALS | 40,581               | 144,973      |

|         |                                    |    |              |  |
|---------|------------------------------------|----|--------------|--|
| SAFETY: | MONTHLY                            |    | YEAR-TO-DATE |  |
|         | First Aid Cases                    | 8  | 50           |  |
|         | Doctor Cases                       | 1  | 7            |  |
|         | Lost Time Accidents                | 0  | 2            |  |
|         | Days since last lost time accident | 54 | 54           |  |

*E. Ray D. Cook*  
Manager of Operations

MONTH: JUNE 1987

# OPERATIONS REPORT - LYONS PLA

MONTHLY

YEAR-TO-DATE

|                                                     |  |  |  |  |                                                 |         |       |           |         |                                                      |         |           |        |     |
|-----------------------------------------------------|--|--|--|--|-------------------------------------------------|---------|-------|-----------|---------|------------------------------------------------------|---------|-----------|--------|-----|
| R. Drill<br>ipping                                  |  |  |  |  | Ft: 3692 Hrs: 89.5 Ft/Hr: 41.3<br>Tons: 130,443 |         |       |           |         | Ft: 51,803 Hrs: 789.0 Ft/Hr: 65.7<br>Tons: 1,039,982 |         |           |        |     |
| SHING:                                              |  |  |  |  | 720                                             |         |       |           |         | 6552                                                 |         |           |        |     |
|                                                     |  |  |  |  | TONS HOURS TONS/HR. % OPER.                     |         |       |           |         | TONS HOURS TONS/HR. % OPER.                          |         |           |        |     |
| Hi. Cal. R#1                                        |  |  |  |  | 34,713                                          | 247.5   | 140.3 | 34.4      | 210,922 | 1489.5                                               | 141.6   | 22.7      |        |     |
| Hi. Cem. R#3                                        |  |  |  |  | 22,584                                          | 161.5   | 139.8 | 22.4      | 89,058  | 634.0                                                | 140.5   | 9.7       |        |     |
| Lo Cem. R#3-4                                       |  |  |  |  | 0                                               |         |       |           | 12,398  | 89.0                                                 | 139.3   | 1.4       |        |     |
| Weathered                                           |  |  |  |  | 10,923                                          | 79.0    | 138.3 | 11.0      | 55,412  | 435.5                                                | 127.2   | 6.6       |        |     |
| R. #4                                               |  |  |  |  | 0                                               |         |       |           | 0       |                                                      |         |           |        |     |
| Silica                                              |  |  |  |  | 2,600                                           | 17.5    | 148.6 | 2.4       | 7,336   | 53.0                                                 | 138.4   | 0.8       |        |     |
| Gypsum                                              |  |  |  |  | 671                                             | 5.0     | 134.2 | 0.7       | 14,983  | 94.5                                                 | 158.6   | 1.4       |        |     |
| Iron                                                |  |  |  |  | 1,170                                           | 8.5     | 137.6 | 1.2       | 5,683   | 43.5                                                 | 130.6   | 0.7       |        |     |
| TOTALS                                              |  |  |  |  | 72,661                                          | 519.0   | 140.0 | 72.1      | 395,792 | 2839.0                                               | 139.4   | 43.3      |        |     |
| RAW GRIND                                           |  |  |  |  | 63,845                                          | 502.5   | 127.1 | 69.8      | 349,525 | 2778.0                                               | 125.8   | 42.4      |        |     |
| KILN                                                |  |  |  |  | 39,023                                          | 680.0   | 57.4  | 94.5      | 201,849 | 3574.5                                               | 56.5    | 54.6      |        |     |
| FINISH MILL:                                        |  |  |  |  |                                                 |         |       |           |         |                                                      |         |           |        |     |
| I                                                   |  |  |  |  | 2,939                                           | 32.0    | 91.8  | 4.4       | 28,248  | 312.5                                                | 90.4    | 4.8       |        |     |
| I LA                                                |  |  |  |  | 0                                               |         |       |           | 0       |                                                      |         |           |        |     |
| II                                                  |  |  |  |  | 7,230                                           | 79.5    | 90.9  | 11.0      | 69,125  | 797.5                                                | 86.7    | 12.2      |        |     |
| II LA                                               |  |  |  |  | 26,099                                          | 303.5   | 86.0  | 42.2      | 122,222 | 1471.0                                               | 83.1    | 22.5      |        |     |
| II MOD                                              |  |  |  |  | 5,110                                           | 56.0    | 91.2  | 7.8       | 24,622  | 278.0                                                | 88.6    | 4.2       |        |     |
| Class G                                             |  |  |  |  | 1,444                                           | 28.5    | 50.7  | 4.0       | 4,921   | 106.0                                                | 46.4    | 1.6       |        |     |
| DUCT                                                |  |  |  |  | 1,119                                           | 20.5    | 54.6  | 2.9       | 13,258  | 253.5                                                | 52.3    | 3.9       |        |     |
| SON                                                 |  |  |  |  | 925                                             | 17.5    | 52.8  | 2.4       | 3,055   | 55.0                                                 | 55.5    | 0.8       |        |     |
| TOTALS                                              |  |  |  |  | 44,866                                          | 537.5   | 83.5  | 74.7      | 265,451 | 3273.5                                               | 81.1    | 50.0      |        |     |
| Equiv. Prod: .85x 39,023 +.15x 44,866 = 39,900 Tons |  |  |  |  | .85x 201,849 +.15x 265,451 = 211,389 Tons       |         |       |           |         |                                                      |         |           |        |     |
| POWER:                                              |  |  |  |  | KWH: 6,507,000 COST/KWH: 0.0468                 |         |       |           |         | KWH: 41,274,000 COST/KWH: 0.0505                     |         |           |        |     |
|                                                     |  |  |  |  | COST: 304,749.15 COST/EQ.TON: 7.64              |         |       |           |         | COST: 2,086,300.86 COST/EQ.TON: 9.87                 |         |           |        |     |
| FUEL:                                               |  |  |  |  | 1022 131,000 12,339                             |         |       |           |         | 1010 131,000 11,967                                  |         |           |        |     |
|                                                     |  |  |  |  | GAS MCF OIL/GAL COAL/TONS BTU/TON               |         |       |           |         | GAS MCF OIL/GAL COAL/TONS BTU/TON                    |         |           |        |     |
| DRYER                                               |  |  |  |  | 7,033                                           | -       | 260   | 187,226   | 29,858  | -                                                    | 1,533   | 168,895   |        |     |
| KILN                                                |  |  |  |  | 1,000                                           | 174,900 | 7,784 | 5,535,900 | 73,957  | 1,534,790                                            | 32,846  | 5,260,815 |        |     |
| TOTALS                                              |  |  |  |  | 8,033                                           | 174,900 | 8,044 | 5,884,515 | 103,815 | 1,534,790                                            | 34,379  | 5,591,990 |        |     |
| MANHOURS:                                           |  |  |  |  | No. Hours Hrs. Pd. Total Man Hrs/               |         |       |           |         | No. Hours Hrs. Pd. Total Man H                       |         |           |        |     |
|                                                     |  |  |  |  | Emp. Worked Not Worked Hours Hours Equiv. T.    |         |       |           |         | Emp. Worked Not Worked Hours Hours Equiv.            |         |           |        |     |
| Hourly                                              |  |  |  |  | 84                                              | 11,200  | 2,002 | 13202     | .33     |                                                      | 113,386 | 16,528    | 129914 | .62 |
| Staff                                               |  |  |  |  | 38                                              | -       | -     | 6585      | .17     |                                                      | -       | -         | 59455  | .28 |
| TOTALS                                              |  |  |  |  | 122                                             | -       | -     | 19787     | .50     |                                                      | -       | -         | 189369 | .90 |
| SALES:                                              |  |  |  |  | Overtime Hours: 314.4                           |         |       |           |         | SAFETY:                                              |         |           |        |     |
|                                                     |  |  |  |  | MONTHLY YEAR-TO-DATE                            |         |       |           |         | MONTHLY YEAR-TO-DATE                                 |         |           |        |     |
| I                                                   |  |  |  |  | 3,768 32,133                                    |         |       |           |         | 6 76                                                 |         |           |        |     |
| I LA                                                |  |  |  |  | 0 0                                             |         |       |           |         | 0 10                                                 |         |           |        |     |
| II                                                  |  |  |  |  | 10,158 84,454                                   |         |       |           |         | 0 3                                                  |         |           |        |     |
| II LA                                               |  |  |  |  | 18,712 111,298                                  |         |       |           |         | 80 80                                                |         |           |        |     |
| "G"                                                 |  |  |  |  | 1,382 2,630                                     |         |       |           |         |                                                      |         |           |        |     |
| MOD                                                 |  |  |  |  | 3,163 20,924                                    |         |       |           |         |                                                      |         |           |        |     |
| DUCT                                                |  |  |  |  | 1,281 13,500                                    |         |       |           |         |                                                      |         |           |        |     |
| IONS                                                |  |  |  |  | 313 3,089                                       |         |       |           |         |                                                      |         |           |        |     |
| cher                                                |  |  |  |  | 6 62                                            |         |       |           |         |                                                      |         |           |        |     |
| TOTALS                                              |  |  |  |  | 38,783 268,090                                  |         |       |           |         | E. D. Cook<br>E. D. Cook, Manager of Operations      |         |           |        |     |
|                                                     |  |  |  |  |                                                 |         |       |           |         | MONTH: SEPTEMBER 1987                                |         |           |        |     |

